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Exploration of students' reaction in medical error events and the impact of personalized training on the speaking-up behavior in medical error events

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ABSTRACT

Background: The ability of medical students to speak up before a medical error occurs is a timely and necessary interaction to prevent potential patient harm. As it may be crucial to improve patient safety, we explored how medical students react to a medical error and provided them appropriate training regarding speaking up about medical issues.

Methods: A quasi-experimental study was conducted in Taiwan involving 153 medical students who participated in a speaking-up simulation course. They were divided into two groups. The first group participated in a non-life-threatening scenario before the intervention, followed by a personalized debriefing session, then a life-threatening scenario after the intervention. The second group participated in a life-threatening scenario before the intervention, followed by a personalized debriefing session, then a non-life-threatening scenario after the intervention. Students also completed patient safety attitude survey.

Results: During the preintervention scenario, the overall medical students' speaking-up rate to medical error was 45.1%. The speaking-up rate of medical students in life-threatening scenario was significantly higher than the rate in non-life-threatening scenario before the intervention (64.6% vs 24.3%, $p < 0.001$). After personalized debriefing, the speaking-up rate to medical errors was significantly improved both in life-threatening scenarios (95.9%, $p < 0.001$) and in non-life-threatening scenarios (100%, $p < 0.001$). Male medical students had significantly higher speaking-up rates than female students in life-threatening scenario (76.2% vs 51.4%, $p = 0.02$). On post-intervention surveys, students provided several reasons for their likelihood of speaking up or remaining silent during a medical error event.

Conclusions: Medical students' rate of speaking-up to medical error was higher in a simulated life-threatening scenario than in a simulated non-life-threatening scenario. Faculty-led personalized debriefing can facilitate medical students' adoption of communication strategies to speak up more in medical error events. Educators should also consider gender differences when they design effective assertive communication courses.

KEYWORDS
Speaking up; assertive communication; medical errors; patient safety; medical students; simulation; personalized; gender differences

Introduction

Patients harmed by medical error is a critical issue in healthcare (Makary and Daniel 2016). Speaking up about a possible medical error in a timely and adequate manner is an effective intervention that can prevent patients from harm (Belyansky et al. 2011; Kolbe et al. 2012). In instances when a healthcare provider is hesitant to speak up or is inadequately assertive in their communication, a patient's safety could be compromised by medical error (Rabøl et al. 2011). According to a national survey conducted in United States, about 75% healthcare providers were concerned about possible errors made by their peers, but only 16% of them had spoken with peers to share their concerns (Maxfield et al. 2005). This situation also appears in Asia; an annual survey conducted by Joint Commission of Taiwan indicated that healthcare providers have concerns about

Practice points

- Personalized assertive communication training could improve medical students speaking up behavior in the event of medical error.
- The medical students have a higher rate of speaking up in the life-threatening scenario compared to the non-life-threatening scenario.
- Male students had a higher rate of speaking up than females in the life-threatening scenario but not in the non-life-threatening scenario.

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speaking up to prevent or report medical error events (Joint commission of Taiwan 2019, 2020). These results indicated that it is crucial to develop and change healthcare providers' behavior to speak up in the event of medical error to improve patient safety (O'Donovan and McAuliffe 2020).

Behavior can be thought of as a chain of conditions beginning with the physical stimulus and ending in the response (Hull 1943). Some studies indicate that interventions to train speaking-up behavior in the event of medical error have less effect on experienced healthcare providers who were in high hierarchical status, compared to less experienced healthcare professionals (Raemer et al. 2016; Omura et al. 2017). This suggests that the development of speaking up behavior in the event of medical error is especially important for those who are in the lowest hierarchy status in the medical team and during an early phase of their healthcare career, such as medical students.

Another questionnaire survey in Taiwan identified some barriers associated with medical students' speaking-up about medical errors such as hierarchy gap, fear of worsening team dynamic, and self-doubt about their own ability to perceive, understand, and communicate issues. Medical students with different gender had different perceived barriers in speaking-up to medical error such as the age gap of senior personnel (Chen et al. 2021). Previous study indicated that gender differences may be a contributing factor which related to different speaking up behavior when witness a medical error event (Pattni et al. 2017). Gender differences also affected the interactions and behaviors within medical team (Etherington et al. 2021). Adequately addressing these barriers and also considering the gender differences would be important to develop effective intervention to train medical students' behavior to speak up about medical errors. Questionnaire surveys could provide useful information to explore these barriers, although bias from recall or social desirability may exist when using self-report questionnaires to collect the healthcare providers' response to medical error events. To date, there is a paucity of studies on this topic using an objective evaluation approach. Most surveys use self-reported questionnaires (Maxfield et al. 2005; Martinez et al. 2017; Joint commission of Taiwan 2019, 2020), or focus groups interviews (Peadon et al. 2020; Levine et al. 2020).

Different pedagogical approaches including simulation, role-play, video presentations, discussion, case studies and workshops have been used to develop healthcare providers' assertive communication competency (Kim et al. 2020; O'Donovan and McAuliffe 2020). Simulation activities provide opportunities for faculty to train and assess students, and for participants to practice and develop speaking-up skills without causing negative consequences to a real patient. In addition, simulation that is grounded by Kolb's experimental learning theory serves as an effective intervention to develop speaking-up behavior in medical error events (Pian-Smith et al. 2009).

In simulation activities, scenarios are the key element that provides concrete experience for the learners and serve as part of the foundation of reflective observation according to Kolb's experimental learning theory. Most studies describe scenarios for training with similar clinical circumstances without varying the severity of the simulated patient (Pian-Smith et al. 2009; Raemer et al. 2016; Omura et al. 2017). Previous research also reveals that male and female students may experience and perceive simulation scenarios differently. These differences may lead to variation in how male and females respond and speak up after witnessing a medical error event. Development of an effective training program requires understanding of the target participants and factors which may affect their learning (Samuel et al. 2012; Chen et al. 2021). There is little research related to "does different severity of medical error scenario affect the response and learning of medical students?" or "does different gender of medical students associate to different response or learning outcomes?".

The aims of the study were to explore (1) The response of medical students speaking up when encountering different medical error circumstances. (2) Understanding the effect of gender differences among medical students when encountering medical error events. (3) The outcomes of a simulation course with personalized debriefing for medical students in speaking-up to medical errors.

Method

Setting and participants

The study was approved by the institutional review board of the Taipei Medical University. The authors conducted a quasi-experiment study at a simulation training center of a medical university in Northern Taiwan from April to November 2018. One hundred fifty-three medical students including 95 male and 58 female medical students participated in this study. All students started with a combined internal medicine and surgery eight-week clerkship rotation, in which patient safety lecture sessions were required for all medical students in the beginning. During the study period, all the participants also participated in the speaking up educational course during a second clinical internal medicine clerkship rotation. The medical students were certified as advance cardiac life support provider prior to enter the simulation course. Participants were divided into two groups: Group one consisted of 74 students who encountered a non-life-threatening simulation scenario before the intervention, followed by personalized debriefing session, then life-threatening simulation scenario after the intervention. Group two consisted of 79 students who encountered life-threatening simulation scenario before the intervention, followed by a personalized debriefing session, then a non-life-threatening simulation scenario after the intervention.

Speaking-up training course

The course design was informed by Kolb's experimental learning theory that posits learning is created by the process of experience transformation, consisting of four key stages: concrete experience, reflective observation, abstract conceptualization and active experimentation (Kolb et al. 2014). Guided by this theory, we designed the simulation course with a preintervention scenario (concrete experience), followed by post-scenario personalized debriefing (reflective observation and abstract conceptualization) and lastly, a postintervention scenario (active experimentation).

In order to assess the students' speaking-up behavior in a medical error event, it was important to make sure they

had sufficient background knowledge to manage the clinical condition that they were about to encounter. Students were informed that the simulation training was to assess and treat the patient with paradoxical supraventricular tachycardia (PSVT) two weeks before the simulation course, and students received pre-course online reading material that included American Heart Association (AHA) Guidelines for cardiopulmonary resuscitation to identify and deal with PSVT patient.

On the course day, students took a pre-simulation test that they had to pass, before participating in the simulation scenario. Faculty assisted students through discussing and instructing them with the course material if they could not pass the pre-simulation test at their first attempt. To avoid their unfamiliarity of the simulation scenario, students were oriented about the simulation setting and their task to manage the simulated patient, but were not informed that there was going to be a medical error event. Immediately after being briefed about the clinical condition of the simulated patient, each student entered the simulation scenario to interact with a full-body computerized mannequin, a standardized nurse, and the standardized senior physician. Students were expected to evaluate and treat the PSVT patient who either had stable vital signs in the non-life-threatening scenario or unstable vital signs in the life-threatening scenario with the standardized nurse. After a student ordered the correct medicine, the standardized nurse prepared it, and then the senior physician would enter the simulation scenario and order an incorrect medication for the patient. If the student did not speak up to the error event, the nurse delivered the incorrect medicine. If the student did speak up to the error event, the senior physician would initiate discussion with the student about their suggestion for treatment option.

After the preintervention scenario, four students participated in a faculty-led personalized debriefing session. In order to standardize the debriefing session, the faculty member followed a 60-min, step-by-step scripted guide. The debriefing process was based on the GAS (gather, analyze, and summarize) model (Phrampus and O'Donnell 2013). The faculty facilitated students' reflection, incorporated a standardized speaking-up technique, and then assisted the students to generate their personalized debriefing strategy and document it in their own words. The students were then able practice these strategies with the group at the end of the debriefing session. The standardized speaking-up technique included two-challenge rules, advocacy and assertion, CUS (concerned, uncomfortable and patient safety issue) and DESC (describe, express, suggest and consequences) script based on TeamSTEPPS 2.0 principles (Pocket Guide: TeamSTEPPS | Agency for Healthcare Research and Quality (ahrq.gov)). After the post-intervention scenario, students and the faculty participated in a second, 30-min, debriefing session to discuss their experience of using their personalized speaking up strategies and to modify them according to their performance in the scenario.

Assessment instruments and data collection

The outcome of this study was to assess the response of students when they encountered a medical error in a simulation scenario. We used a modified Pian-Smith Evaluation Tool to measure speaking up. Students were assessed on a 5-point scale: 1 = no reaction; 2 = negative facial or body language expression; 3 = says something oblique, obtuse; 4 = advocates or inquires; 5,=advocates or inquires repeatedly, initiates discussion (Oner et al. 2018). Our expert panel including one simulation expert, one senior clinical physician, one senior practicing nurse and one nurse practitioner defined that a score equal or greater than 4 was a clear speaking-up behavior to medical error. Through standardized videos and discussion before the course, two faculties were trained as the independent raters to assess the student's speaking up behavior in real-time scenario. One served as the standardized nurse, and one as the standardized senior physician. The raters were aware of the pre- and post-intervention status of the students. If the raters had concerns regarding the students' speaking-up behavior, the rater would discuss with the senior physician and standardized nurse to reach a consensus as the result. The raters observed and assessed students' speaking-up behavior in the simulated medical error scenario by a checklist with a dichotomous scale with 1 (speaking up) and 0 (not speaking up). Successfully speaking up to medical error event was defined as speaking up before the standardized nurse delivered the incorrect medication ordered by standardized senior physician. The speaking-up behavior of the students in the medical error event was assessed during the preintervention and postintervention scenarios. According to Krippendorff's alpha, the inter-rater reliability was 0.842 in the non-life-threatening scenario and 0.952 in the life-threatening scenario, respectively, indicating a high level of agreement (Zapf et al. 2016).

A 19-item self-reported patient safety attitude questionnaire was adopted based on the Safety Attitudes Questionnaire (Sexton et al. 2006). The study selected items related to teamwork and safety climate from the original questionnaire that were suitable for medical students. Safety attitude was assessed on a 6-point scale from 1 (strongly disagree) to 6 (strongly agree) before and after the intervention.

Students' reactions to the medical error scenario were collected to facilitate personalized debriefing session. At the beginning of each personalized debriefing session, faculty reviewed students' reaction when encountering the medical error in the simulation scenario and explored the underlying reasons of their actions. Faculty recorded the students' responses on paper during the debriefing session and then categorized the data into four categories including: (1) the reasons of speaking-up to medical error in the life-threatening scenario; (2) the reasons of not speaking-up to medical error in the life-threatening scenario; (3) the reasons of speaking-up to medical error in the non-life-threatening scenario, and; (4) the reasons of not speaking-up to medical error in the non-life-threatening scenario.

Data analysis

We analyzed data using SPSS version 19.0 (IBM Corp., Armonk, NY) for Microsoft windows. We report descriptive statistics for age, gender, and experiences of patient safety

Table 1. Demographic characteristic between groups ($N = 153$).

Variable	Group 1 $n = 74$	Group 2 $n = 79$	Total	p
Age (years), Mean (range)	23.84 (22–33)	23.69 (21–32)		0.97[a]
Gender n (%)				
Male	53 (71.6)	42 (53.2)	95	0.02*
Female	21 (28.4)	37 (46.8)	58	
Experience of clerkship rotation (weeks), Mean (range)	26.07 (18–34)	26.05 (18–34)		0.97
Experience of clerkship rotation of internal medicine and surgery, Yes, n (%)	74 (100%)	79 (100%)	153	
Experience of patient safety training, Yes, n (%)	74 (100%)	79 (100%)	153	
Experience of speaking up to medical error[b], Yes, n (%)	35 (47.3)	38 (49.4)		0.87
Baseline PSA score, Mean (range)	87.73 (71-104)	88.11 (70-107)		0.89[a]

Note. Group 1: encountering Non-life threatening scenario first; Group 2: encountering life threatening scenario first; PSA: patient safety attitude; Chi-Square Test; a: Mann-Whitney U test; b: two students did not complete the survey in group 2.

Table 2. Speaking-up rate to medical error and patient safety attitude among before and after personalized debriefing.

Outcome	Before Debriefing	After Debriefing	Crosstabs *Chi-square*	p
Speaking up to medical error n (%)				
Overall				
Life-threatening	51 (64.6%)	71 (95.9%)	23.301	<0.001[a]
Non-life-threatening	18 (24.3%)	79 (100%)	94.298	<0.001[a]
Male				
Life-threatening	32 (76.2%)	52 (98.1%)	10.999	<0.002[a]
Non-life-threatening	11 (20.8%)	42 (100.0%)	59.658	<0.001[a]
Female				
Life-threatening	19 (51.4%)	19 (90.5%)	9.077	<0.004[a]
Non-life-threatening	7 (33.3%)	37 (100.0%)	32.515	<0.001[a]
PSA ($n = 146$) , Mean (*SD*)	87.92 (6.64)	89.72 (7.02)		0.001

Note. PSA: patient safety attitude; [a]Fisher's exam; $^*p < 0.05$, two-tailed.

training. Age is presented in mean with range (minimum to maximum), while experiences of patient safety training are shown in the number of events. Chi-square test was applied to examine the difference in the proportion of speaking-up rate between scenarios (life-threatening versus non-life-threatening), tests (pre-test and post-test), as well as gender (male and female). Statistical significance was based on common threshold p-value 0.05.

Content analysis of the qualitative data was conducted by two independent coders. The coders identified similar sentences of the reaction reasons within the same category and then integrated them into one sentence which could contain those reasons. Subsequently, they discussed and reached consensus for the final result.

Results

One hundred fifty-three medical students with a mean age of 23.8 years (ranged from 21 to 33 years) completed the simulation course, including 95 male students and 58 female students. There were no significant differences between the two groups in experience of clerkship rotation, patient safety training, and speaking up to medical error. Seven students did not finish the safety attitude questionnaire before the intervention. There is no statistically significant difference of patient attitude scores between the two groups before the intervention. Overall demographic information of the participants is presented in Table 1.

Experience of speaking up to medical error in the two groups were 47.3% and 49.4% in Table 1. The overall rate of medical students' speaking up to medical error was 45.1% before the intervention (Appendix Table A1). The rate of medical students' speaking up to medical error in the life-threatening scenario was significantly higher than the rate in the non-life-threatening scenario before the intervention (64.6% vs 24.3%, $p < 0.001$). The results also showed that before personalized debriefing, male students had a higher rate of speaking up than females in the life-threatening scenario (76.2% vs 51.4%, $p < 0.02$), but there were no differences between genders in non-life-threatening scenario (Appendix Table A2).

After personalized debriefing, the speaking-up rate of medical students in medical error scenario significantly improved in life-threatening scenarios (from 64.6% to 95.9%; $p < 0.001$) and non-life-threatening scenarios (from 24.3% to 100%; $p < 0.001$). Personalized debriefing also significantly improved the speaking-up behavior in a medical error event for male and female medical students in the non-life-threatening and life-threatening scenario (Male students in life-threatening scenario 76.2% → 98.1%, $p < 0.002$; in non-life-threatening scenario 20.8% → 100%, $p < 0.001$; female students in life-threatening scenario 51.4% → 90.5%, $p < 0.004$; in non-life-threatening scenario 33.3% → 100%, $p < 0.001$) at postintervention scenario (Table 2). Students' patient safety attitude was significantly improved from mean 87.9 to 89.7 (Wilcoxon test, $z = -3.33$; $p = 0.001$) after the intervention.

The free response data revealed that medical students' common the reasons of speaking-up to medical error included "The patient's life is in danger so I have to react," "As a doctor, I should be able to provide the most appropriate treatment for the critically ill patient," and "I should provide my opinions to assist the medical team in this kind of critically medical condition." The medical students' common the reasons of not speaking-up to medical error including "Senior physician should be in charge in this kind of critical condition," "I did not have enough clinical experience and may delay the treatment the critically ill patient," and "If I was wrong, the patient may get worse and others

Table 3. Reasons for speaking-up or not speaking up to medical errors.

Reasons for speaking-up to medical error in life-threatening scenario
a. The patient's life is in danger so I have to react.
b. The patient is in critical condition and there is no room for mistake.
c. As a doctor, I should be able to provide the most appropriate treatment for the critically ill patient.
d. I should provide my opinions to assist the medical team in this kind of critically medical condition.
e. I would raise my concerns when other's opinion was different than mine.
Reasons for speaking-up to medical error in non-life-threatening scenario
a. I wanted to make confirmation about my thoughts and the underlying reasons of the different management.
b. This is not a life-threatening condition so there is time for discussion.
c. I would raise my concerns when other's opinion was different than mine.
Reasons for speaking-up to medical error in life-threatening scenario
a. I did not have enough clinical experience and may delay the treatment the critically ill patient.
b. Senior physician should be in charge in this kind of critical condition.
c. If I was wrong, the patient may get worse and others in the medical team would have negative impression on me or blame me.
d. This is the senior physician's decision, not my concern.
e. If senior physician was wrong, nurse would speak up for the wrong order from senior physician because she had more experience.
Reasons for speaking-up to medical error in non-life-threatening scenario
a. I would have more confidence to provide my suggestion if the senior physician's order was not effective.
b. There may be other treatment which I never learned before. I could discuss with the senior physician that after treating this patient.
c. The patient's life is not in danger. I could provide my suggestion after the senior physical managed the patient.

in the medical team would have negative impression on me or blame me." (Table 3)

Discussion

The results of this study indicate that the students' speaking-up rate in medical error events is significantly higher in life-threatening scenarios than in non-life-threatening scenarios. Personalized debriefing was also shown to be an effective intervention that could significantly improve medical students' speaking-up behavior during medical error events in life-threatening and non-life-threatening scenarios, and in increasing students' patient safety attitude. Interestingly, male medical students had significantly higher speaking-up rate in medical error events than female medical students in life-threatening scenarios, but not in non-life-threatening scenarios.

This study contributes to the current literature regarding how students respond to simulated medical error scenarios based on differing clinical severity of the patient, and how gender differences affect students' speaking up behavior in simulated medical error scenarios. The results of this study indicate that personalized debriefing can facilitate medical students' ability to develop their own speaking up strategies and improve their performance when encountering a simulated medical error event. This finding differs from previous studies which found that speaking up training had no effect on improving the ability of practicing clinicians (Raemer et al. 2016; Oner et al. 2018). This may highlight the importance of early educational intervention of speaking up to a medical error event for those who were in their early phase of healthcare career. Previous studies have reported that factors affecting healthcare providers' behavior of speaking up to medical error events including a sense of responsibility toward patients, responsibility as professionals, motivation to speak up to help the patient (such as the perceived risk for patients), the ambiguity or clarity of the clinical situation, confidence based on experience, communication skills, educational background, and concerns over appearing incompetent (Maxfield et al. 2005; Lyndon et al. 2012; Okuyama et al. 2014). Similar findings were made during personalized debriefing sessions in this study. Students who spoke up when a medical error occurred in patients with life-threatening conditions shared with faculty that they had a responsibility toward the patient, that it was life-saving situation, or that they wanted to assist the medical team. For those students who did not speak up when a medical error occurred in patients with life-threatening conditions, they indicated that they thought senior practitioners should be responsible in this critical scenario, they respected the seniors' decision, or their level of confidence decreased when senior personnel had a different medical decision from theirs.

For healthcare providers, speaking up to medical errors is a complex behavior related to cultural and institutional background. Barriers, including fear of retribution, professional hierarchy, personal capability, interpersonal factors, and lack of training could impede healthcare providers speaking up to medical errors (Omura et al. 2017; Peadon et al. 2020). Medical students' willingness to speak up to inadequate hand hygiene decreases when the hierarchy gap increases (Samuel et al. 2012). Medical students, one of the lowest ranks of the health care system, have their own concerns in speaking up to medical error events including potential penalty, compromising the relationship with senior personnel, lack of knowledge and communication skills, hierarchy gap, and the personality of the senior personnel (Chen et al. 2021). Understanding medical students' behavior of speaking up to medical errors in different scenarios is important because educators can use this information to integrate and develop effective assertive communication training courses.

Development of positive speaking-up behavior in medical error events is important for junior healthcare personnel. Even when medical students perceived a patient safety concern that was related to a medical error, only 44% of them spoke up in a critical clinical situation (Schwappach et al. 2019). Our findings similarly show that only 45% of medical students spoke up to medical error in simulation scenario before intervention. It is challenging for students to learn the concept of patient safety during clinical rotations, and disclosure of medical error is not easy. This is because a positive response culture is not well established in most healthcare work environments (Lee et al. 2018). Effective intervention is essential for medical students to develop desirable speaking-up behavior in medical error events. Encouraging healthcare providers to speak up in the event of medical error is related to higher speaking up

frequency and lower withholding voice frequency (Schwappach and Richard 2018). It is important for institutions and their faculty to develop and foster an encouraging environment and to encourage medical students to speak up in the event of medical error in training activities.

Developing personalized strategy is one of the keys for medical students to adopt the positive speaking-up behavior in the event of medical error. There are some practical communication tools that can be used as guides to train others to speak up in the event of medical error. These include the two challenge rule, CUS and DESC from TeamSTEPPS, PACE (Probe, Alert, Challenge and Emergency action) (Nickson 2020), CREC (Conclusion, Reason, Example and Conclusion) (Omura et al. 2019), and 5-step advocacy (attention getter, state your concern, state the problem, state a solution and obtain agreement) (Brindley and Reynolds 2011). However, medical students have to adopt these communication tools into their own language so that they would be more likely to speak up to medical error in the future. Students reacted differently in different types of scenarios. Faculty-led personalized debriefing facilitates students' reflection on their behavior in the scenario and building up different strategies. Students may speak up in the event of medical error more or less depending on the clinical scenario. Furthermore, it is important to consider cultural differences to maximize medical students' benefit from debriefing sessions (Chung et al. 2013). While the results of this study indicate that medical students' patient safety attitude improvement was statistically significant after the intervention, the score of improvement is limited. The very small improvement might be a result of two potential reasons: (1) these participants had received training of patient safety in orientation program when they initiated their clinical course; and (2) this educational course was designed to develop students' speaking up behavior in the specific event of medical error as opposed to a comprehensive patient safety program. A comprehensive educational program related to patient safety topics is essential for developing medical students' patient safety attitude. Future studies should further focus on developing a more comprehensive patient safety educational program to not only develop the participants' speaking up behavior in a medical error event, but also their overall attitude to improve and maintain patient safety.

There is little previous literature that addresses the relationship between gender differences and the behavior of speaking up in medical error events. A recent study from our research team indicated that the factors related to speak up to medical error differ among male and female students such as age gap of the senior personnel (Chen et al. 2021). The study indicates that the speaking-up rate to medical error is significantly higher in life-threatening scenario among male students compared to female students. However, in a survey conducted at a medical school in Australia, female medical students were more likely than male students to speak up for inadequate hand hygiene (Samuel et al. 2012). In addition to gender alone, cultural factors including high power distance, collectivism, and a male-biased medical hierarchy could compromise the team communication (Perry et al. 2018). When building up effective learner-centered educational activity to develop speaking-up behavior in the event of medical error, educators should consider the type of clinical scenario, learner culture, and gender differences, and their effect on learning outcome.

Limitations

There are several limitations in this study. This study was conducted at a single institution in Taiwan, and the results may not be generalizable to other institutions and cultures. Medical students' speaking-up behavior in medical error events was recorded in a simulation scenario. The speaking-up behavior of the students may be different when it is in real clinical setting. Our participants were medical students, who were relatively new to their medical careers. Further follow up study is needed to understand the retention of the learning effect.

Conclusions

Medical students speak up more about possible medical errors during a simulated life-threatening scenario than in a simulated non-life-threatening scenario. Of the medical students who speak up, male students speak up more often in life-threatening situations. Personalized debriefing facilitates medical students' adoption of communication strategies and improves their speaking up performance in the event of medical error. Programs that develop speaking up behavior for students who are in their early healthcare career may have positive effects. Learner culture and gender differences should be taken into consideration when educators develop an assertive communication course.

Disclosure statement

Yi-Chun Chen, S. Barry Issenberg, Yu-Jui Chiu, Hui-Wen Chen, Zachary Issenberg, Yi-No Kang, Che-Wei Lin and Jen-Chieh Wu declare that they have nothing to disclose regarding financial or non-financial conflicts of interest to this manuscript.

Glossary

Assertive communication: Is "the ability to speak and interact in a manner that considers and respects the rights and opinions of others while also standing up for your rights, needs, and personal boundaries".

Speaking up: Is one form of assertive communication and refers to communicating specific observations, requesting clarification or challenging the decision of someone with positional power or authority to act.

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